

School of Computer Science and Artificial Intelligence

Lab Assignment # 9.2

Program	: B. Tech (CSE)
Specialization	:
Course Title	: AI Assisted Coding
Course Code	: 23CS002PC304
Semester	II
Academic Session	: 2025-2026
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Batch No	: 51
Date	: 06/02/26

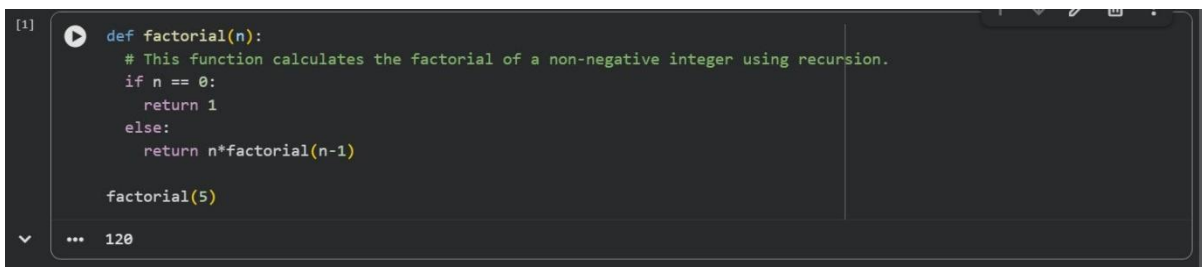
Task Description -1 (Documentation – Function Summary Generation)

Code used:

```
def factorial(n):  
    if n == 0:  
        return 1  
    else:  
        return n*factorial(n-1)  
factorial(5)
```

Prompt: Add a short summary describing the purpose of the function in the code

Output:



```
[1] def factorial(n):  
    # This function calculates the factorial of a non-negative integer using recursion.  
    if n == 0:  
        return 1  
    else:  
        return n*factorial(n-1)  
  
    factorial(5)  
... 120
```

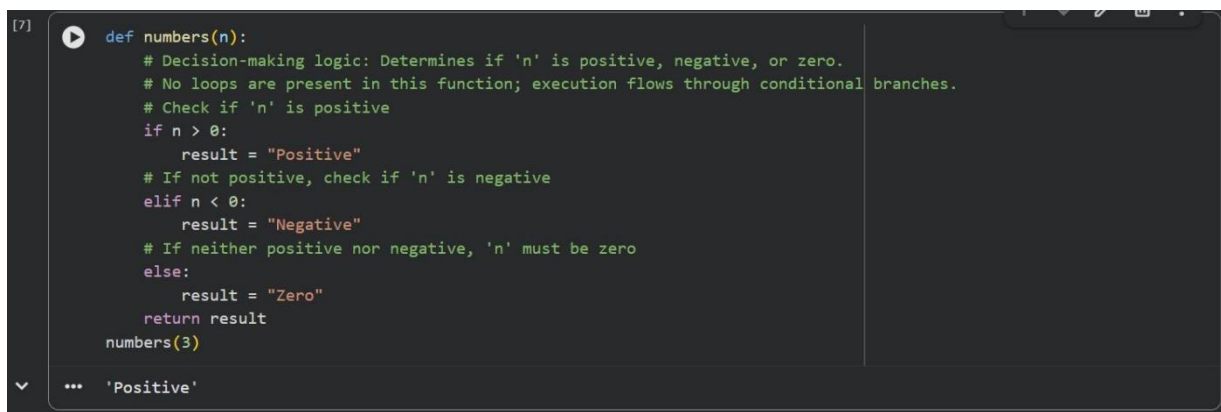
Task Description -2 (Documentation – Logical Explanation for Conditions and Loops)

Code Used:

```
def numbers(n):  
    if n > 0:  
        result = "Positive"  
    elif n < 0:  
        result = "Negative"  
    else:  
        result = "Zero"  
    return result  
numbers(3)
```

Prompt: explain only decision-making logic and loop behavior inside the code

Output:



```
[7]: def numbers(n):  
    # Decision-making logic: Determines if 'n' is positive, negative, or zero.  
    # No loops are present in this function; execution flows through conditional branches.  
    # Check if 'n' is positive  
    if n > 0:  
        result = "Positive"  
    # If not positive, check if 'n' is negative  
    elif n < 0:  
        result = "Negative"  
    # If neither positive nor negative, 'n' must be zero  
    else:  
        result = "Zero"  
    return result  
    numbers(3)  
  
... 'Positive'
```

Task Description -3 (Documentation – File-Level Overview)

Code Used:

```
def factorial(n):  
    result = 1  
    for i in range(1, n + 1):  
        result *= i  
    return result  
  
def square_list(numbers):  
    return [n * n for n in numbers]  
  
def main():  
    print(factorial(5))  
    print(square_list([1, 2, 3, 4]))  
  
main()
```

Prompt: Generate a high-level overview summarizing the overall purpose and functionality of the Code. add the summary at the beginning

Output:

```
[8]  
✓ 0s # This code defines two utility functions: `factorial` to calculate the factorial of a number,  
# and `square_list` to square each element in a list. It then demonstrates their usage in a `main` function.  
  
def factorial(n):  
    result = 1  
    for i in range(1, n + 1):  
        result *= i  
    return result  
  
def square_list(numbers):  
    return [n * n for n in numbers]  
  
def main():  
    print(factorial(5))  
    print(square_list([1, 2, 3, 4]))  
  
main()  
  
... 120  
[1, 4, 9, 16]
```

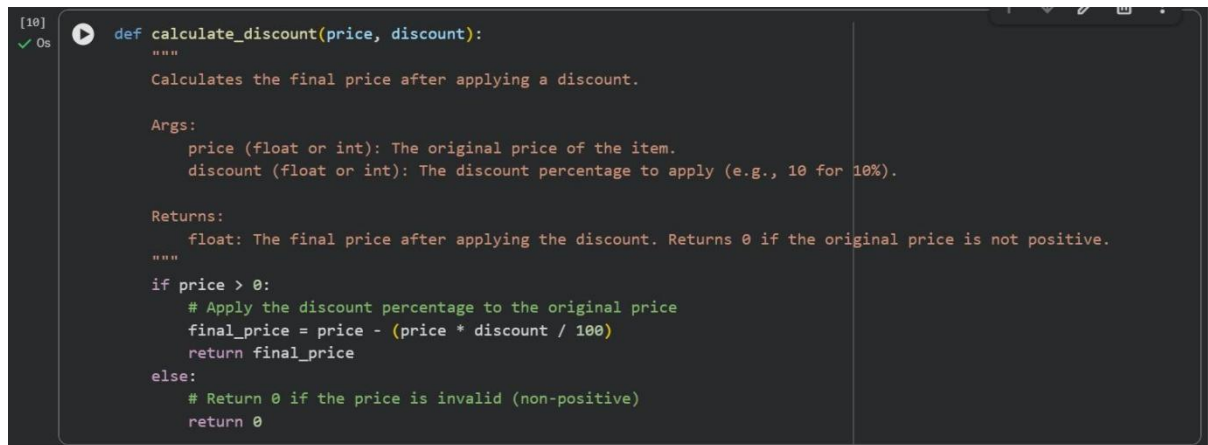
Task Description -4 (Documentation – Refine Existing Documentation)

Code Used:

```
def calculate_discount(price, discount):  
    # calculate  
    if price > 0:  
        # apply discount  
        final_price = price - (price * discount / 100)  
        return final_price  
    else:  
        # wrong price  
        return 0
```

Prompt: Rewrite the documentation to improve clarity and consistency.

Output:



```
[10]  
✓ 0s def calculate_discount(price, discount):  
    """  
    Calculates the final price after applying a discount.  
  
    Args:  
        price (float or int): The original price of the item.  
        discount (float or int): The discount percentage to apply (e.g., 10 for 10%).  
  
    Returns:  
        float: The final price after applying the discount. Returns 0 if the original price is not positive.  
    """  
    if price > 0:  
        # Apply the discount percentage to the original price  
        final_price = price - (price * discount / 100)  
        return final_price  
    else:  
        # Return 0 if the price is invalid (non-positive)  
        return 0
```

Task Description -5 (Documentation – Prompt Detail Impact Study)

Code Used:

```
def reverse_string(text):  
    return text[::-1]
```

Prompt - 1: Write a short documentation comment explaining what this Python function does.

Output - 1:

```
[12]  
✓ Os  def reverse_string(text):  
    """  
    Reverses a given string.  
  
    Args:  
        text (str): The input string to be reversed.  
  
    Returns:  
        str: The reversed string.  
    """  
    return text[::-1]
```

Prompt - 2: Write a clear and detailed documentation comment for the given Python function.

Output - 2:

```
[13]  
✓ Os  def reverse_string(text):  
    """  
    Reverses the given input string.  
  
    This function takes a string as input and returns a new string  
    with the characters in reverse order. It utilizes Python's string slicing  
    feature to achieve this efficiently.  
  
    Args:  
        text (str): The input string that needs to be reversed.  
  
    Returns:  
        str: A new string containing the reversed characters of the input string.  
  
    Examples:  
        >>> reverse_string("hello")  
        'olleh'  
        >>> reverse_string("Python")  
        'nohtyP'  
        >>> reverse_string("12345")  
        '54321'  
    """  
    return text[::-1]
```